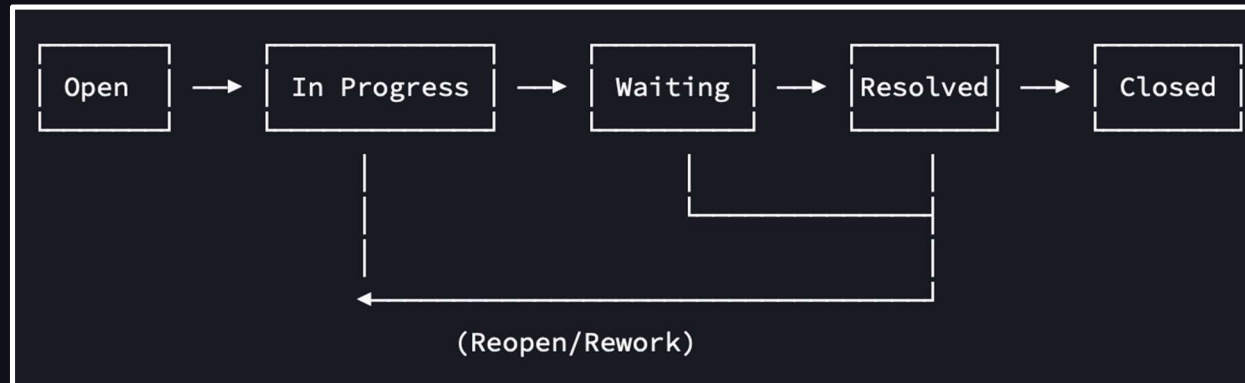


Help Desk Operations

Employee Performance Analytics
with Real World Data

CONTEXT

A help desk is a structured support function where tickets (or issues) are created, triaged, worked on by support employees, and eventually closed. Tickets move through a workflow and can be handed over between employees if needed (so-called turns).



Approach:

Using process and internal communication data to quantify turn-level performance and SLA risk, thereby evaluating manager ratings (Q-Scores) with operational and more objective proxies (O-Scores).

RELEVANCE

Reality:

- Ticket queues fluctuate (peaks, outages, recurring incidents).
 - A single ticket can require multiple interactions and multiple assignees.
 - Customer experience depends on speed, clarity, and correctness.
- Help desks are measured by efficiency (time/cost) and service quality.

Goal:

- Evaluate performance using employee/turn using ticket process & communication data
- Compare performance scores (manager-ratings & operational scores)
- Predict performance scores through key drivers (features)

Business value:

- Identify training needs and highlight coaching targets
- Improve service consistency and customer experience
- Enable monitoring/alerts and process improvements

DATA

- real world data (doi: 10.17632/btm76zndnt.2)
- International help desk software company
- Jan 2016 – Mar 2023

issues.csv
(ticket level):

- priority,
- type,
- project,
- timestamps,
- workflow aggregate
(wf_*, wfe_*,
wf_total_time)

issues_snapshot.csv
(turn level):

splits tickets into
assignee turns
(started, ended, turn)
and allocates workflow
time per assignee

issues_snapshot_
sample.xlsx
(labeled sample):

- manager ratings
(Q1/Q2/Q3)
- spent hours
- steps
- comments,
- validity flag
- notes

sample_utterances.csv
(utterance level):

communication signals
(roles, privacy flags,
masked text)

issues_change_history
.csv (event log):

assignee/status
changes over time

PERFORMANCE SCORES

Q-SCORES:

- Q1 – Quality of Work: Accuracy & precision
 - Q2 – Quality of Work: Thoroughness & completeness
 - Q3 – Client Relations: Responsiveness & courteousness
- Manager Ratings (subjective)

O-SCORES:

- Quality (35%): reopen rate, success rate
 - Efficiency (25%): processing time
 - Productivity (20%): ticket volume, processing steps
 - Communication (20%): first-touch rate, comment activity
- Operational Score (objective)